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Sub ject : DSA

Unit 2, S4, SLO2 - Array Implementation of Lists - Programming Practice 2

1. Find the sum and average of the elements in an array.

Answer: #include <stdio.h>
int main() {
int a[] = {10, 20, 30, 40, 50}, n = 5, sum = 0;
for(int i = 0; i < n; i++) sum += a[i];
printf("Sum = %d, Average = %.2f\n", sum, (float)sum / n);
return 0;
}
Output: Sum = 150, Average = 30.00

2. Given a set of elements find the Armstrong numbers.

Answer: #include <stdio.h>
#include <math.h>
int isArmstrong(int num) {
int temp = num, sum = 0, digits = 0;
while(temp) { digits++; temp /= 10; }
temp = num;
while(temp) { sum += pow(temp % 10, digits); temp /= 10; }
return sum == num;
}
int main() {
int a[] = {153, 120, 370, 371, 407, 500}, n = 6;
for(int i=0; i<n; i++) if(isArmstrong(a[i])) printf("%d is Armstrong\n", a[i]);
return 0;
}

3. Decimal to octal conversion.

Answer: #include <stdio.h>
void decToOctal(int n) {
int octal[32], i = 0;
while(n != 0) { octal[i++] = n % 8; n /= 8; }
printf("Octal: "); for(int j = i - 1; j >= 0; j--) printf("%d", octal[j]); printf("\n");
}
int main() { decToOctal(100); return 0; }
Output: Octal: 144

4. Sum of even numbers in an array.

Answer: #include <stdio.h>
int main() {
int a[] = {1, 2, 3, 4, 5, 6, 7, 8}, n = 8, sum = 0;
for(int i = 0; i < n; i++) if(a[i] % 2 == 0) sum += a[i];
printf("Sum of even numbers = %d\n", sum);
return 0;
}

Output: Sum of even numbers = 20

5. Search an element in an array.

Answer: #include <stdio.h>

```
int main() {  
    int a[] = {15, 32, 47, 58, 64}, n = 5, key = 47, found = -1;  
    for(int i = 0; i < n; i++) if(a[i] == key) { found = i; break; }  
    if(found != -1) printf("Element %d found at index %d\n", key, found);  
    else printf("Not found\n");  
    return 0;  
}
```

Output: Element 47 found at index 2